BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 4197 / 4196

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q1. Q. 1

Q1. If f(x) = e^√(x−1) , then f'(0) =

(A) (B)

(C) (D)

Q2. $\int e^x(\sin x + \cos x)dx$ =

(A) (B)

(C) (D)

Q3. $\int dx/(x(\ln 2x)^3)$ =

(A) (B)

(C) (D)

Q4. If $f(x) = x^2$, then range of f is

(A) (B)

(C) (D)

Q5. If f(x) = x sec x , then f(π) =

(A) (B)

(C) (D)

Q6. If $y = e^{-ax}$, then y dy/dx =

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp partially covers Q6; the question and all four options are fully readable.

Q7. $f(x) = 4 - x^2$ decreases in the interval

(A) (B)

(C) (D)

Q8. $1/(1 + x^2)$ is the derivative of

(A) (B)

(C) (D)

Q9. A vector perpendicular to both $2\hat{i}$ and $\hat{k}$ is

(A) (B)

(C) (D)

Q10. The angle between the vectors $2\hat{i} + 3\hat{j} + \hat{k}$ and $2\hat{i} - \hat{j} - \hat{k}$ is

(A) (B)

(C) (D)

Q11. $\int_0^{(\pi/4)} \sec^2 x \, dx =$

(A) (B)

(C) (D)

Q12. $\int_{-1}^{3} x^3 \, dx =$

(A) (B)

(C) (D)

Q13. The lines represented by $ax^2 + 2hxy + by^2 = 0$ are perpendicular if

(A) (B)

(C) (D)

Q14. The equation of y-axis is

(A) (B)

(C) (D)

Q15. Slope of the line perpendicular to $3x - 4y + 5 = 0$ is

(A) (B)

(C) (D)

Q16. The graph of the Inequality $y < b$ is a / an

(A) (B)

(C) (D)

Q17. Angle Inscribed in a semi-circle is

(A) (B)

(C) (D)

Q18. Equation of normal to the circle $x^2 + y^2 = 25$ at point (4 , 3) is

(A) (B)

(C) (D)

Q19. If $c = \sqrt{65}$, $b = 7$ and $a = 4$, then eccentricity of hyperbola is

(A) (B)

(C) (D)

Q20. If P(2,3) and Q(6,−2) are two points in the plane, then vector $\vec{PQ}$ is

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) Define exponential function.

(ii) Prove the identity $\sec h^2 x = 1 - \tanh^2 x$

(iii) For real valued functions f and defined as $f(x) = 3x^4 - 2x^2$, $g(x) = 2/\sqrt{x}$, $x \neq 0$ Find fog (x) and gof (x)

(iv) Evaluate the limit by algebraic techniques $\lim(h \to 0) (\sqrt{(x+h)} - \sqrt{x})/h$

(v) Find by definition, the derivative of x^(2/3) with respect to 'x'

(vi) Differentiate with respect to x of $((x^2 + 1)^2)/(x^2 - 1)$

(vii) Find dy/dx if $x^2 - 4xy - 5y = 0$

(viii) Differentiate with respect to 'θ' of $\tan^3 \theta \sec^2 \theta$

(ix) Find dy/dx if $y = x^2 \ln \sqrt{x}$

(x) Find y_4 if $y = \sin 3x$

(xi) Prove that e^(x+h) = $e^x\{1 + h + h^2/2! + h^3/3! + ...\}$

(xii) Find interval in which 'f ' is increasing or decreasing if $f(x) = x^2 + 3x + 2$, $x \in (-4 , 1)$

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Using differentials, find dx/dy when $xy - \ln x = c$

(ii) Evaluate $\int_1^3 (3 - \cos 2x)/(1 + \cos 2x)\, dx$

(iii) Find the area between the x − axis and the curve y = sin 2x from x = 0 to x = π/3

(iv) Solve the differential equation sec x + tan y dy/dx = 0

(v) Evaluate $\int_0^{(\pi/6)} x \cos x \, dx$

(vi) Evaluate $\int x^2 \ln x \, dx$

(vii) Find $\int x^2/(4 + x^2) \, dx$

(viii) Find the point three fifth of the way along the line-segment from A(-5 , 8) to B(5 , 3).

(ix) Write down an equation of straight line passing through (5 , 1) and parallel to line passing through points (0 , -1) , (7 , -15)

(x) The xy – coordinate axes are translated through point O' whose coordinates are given in xy – coordinate system. The coordinates of P are given in XY-coordinate system. Find coordinates of P in xy – coordinate system, here P(-5 , -3) , O'(-2 , -6).

(xi) Find area of the triangular region whose vertices are A(5 , 3) , B(-2 , 2) , C(4 , 2).

(xii) Find an equation of each of the lines represented by $10x^2 - 23xy - 5y^2 = 0$

Q4. (Marks 9 × 2 = 18) 4. Answer briefly any Nine parts from the followings:-

(i) What is an objective function?

(ii) Graph the solution set of 3x − 2y ≥ 6

(iii) Find centre and radius of circle $x^2 + y^2 + 12x - 10y = 0$

(iv) Write an equation of the circle with centre (-3 , 5) and radius 7.

(v) Find the focus and directrix of parabola $x^2 = 4(y - 1)$

(vi) Find the focus and vertex of parabola $y = 6x^2 - 1$

(vii) Find the foci and vertices of ellipse $9x^2 + y^2 = 18$

(viii) Find the eccentricity of hyperbola $25x^2 - 16y^2 = 400$

(ix) Find the direction cosines of vector v = 6i − 2j + k

(x) Find 'α' so that |αi + (α + 1)j + 2k| = 3

(xi) Calculate the projection of a = 3i + j − k along b = −2i − j + k

(xii) Prove that a × (b + c) + b × (c + a) + c × (a + b) = 0

(xiii) Find the value of α , so that αi + j, i + j + 3k and 2i + j − 2k are coplaner.

Q. Section ---------- II

Note: Attempt any three questions. (10 × 3 = 30)

Q5. 5 -(a) Evaluate the limit $\lim_{\theta \to 0} (\tan \theta - \sin \theta)/\sin^3 \theta$

(b) If $y = \sqrt{(\tan x + \sqrt{(\tan x + \sqrt{(\tan x + ...\infty)})})}$, Prove that $(2y - 1)\, dy/dx = \sec^2 x$

Q6. 6 -(a) Show that $Y = X^x$ has minimum value at $X = 1/e$

(b) Evaluate $\int x/(x^4 + 2x^2 + 5)\, dx$

Q7. 7 -(a) Evaluate $\int_0^{(\pi/4)} \cos^4 t\, dt$

(b) Maximize $f(x,y) = 2x + 5y$ subject to the constraints $2y - x \leq 8$; $x - y \leq 4$; $x \geq 0$; $y \geq 0$

Q8. 8 -(a) Find an equation of a circle of radius 'a' and lying in the second quadrant such that it is tangent to both the axes.

(b) Prove that the line segments joining the mid points of the sides of a quadrilateral taken in order form a parallelogram.

Q9. 9 -(a) Find centre, foci and directrices of the ellipse $x^2 + 16x + 4y^2 - 16y + 76 = 0$

(b) Find a joint equation of lines through the origin and perpendicular to the lines $x^2 - 2xy \tan \alpha - y^2 = 0$

GROUP – II

Q1. Q. 1

Q1. $\int \ln a \cdot a^x\, dx =$

(A) (B)

(C) (D)

Q2. $\int e^x/(e^x - 1)\, dx =$

(A) (B)

(C) (D)

Q3. $\lim(x\to 0)\ (1 + 3x)^{(2/x)} =$

(A) (B)

(C) (D)

Q4. The perimeter P of a square as a function of its area A is

(A) (B)

(C) (D)

Q5. If $f(x) = \cot x$ then $f'(\pi/6) =$

(A) (B)

(C) (D)

> Note: A 'BLANK' cancellation stamp fully covers Q5 option (B); marked [illegible in source scan].

Q6. $d/dx\ [\ln(e^x + e^{-x})] =$

(A) (B)

(C) (D)

Q7. If $y = \text{Sinh}^{-1}(x^3)$ then $dy/dx =$

(A) (B)

(C) (D)

Q8. The derivative of $y = \sec^{-1} x/a$ is

(A) (B)

(C) (D)

Q9. The lines joining the mid points of any two sides of a triangle is always _____ to the third side.

(A) (B)

(C) (D)

Q10. If u and v be any vectors, then $u \times v$ is

(A) (B)

(C) (D)

Q11. $\int_a^b f(x)\ dx =$

(A) (B)

(C) (D)

Q12. $\int_0^4 x\ dx =$

(A) (B)

(C) (D)

Q13. The slope of the line $2x + 3y - 1 = 0$ is

(A) (B)

(C) (D)

Q14. The lines lying in the same plane are called

(A) (B)

(C) (D)

Q15. If the points (a , 0) , (0 , b) and (x , y) are collinear then

(A) (B)

(C) (D)

Q16. The graph of $x + 2y \leq 6$ is

(A) (B)

(C) (D)

Q17. The fixed line of the conic is known as

(A) (B)

(C) (D)

Q18. The equation $a(x^2 + y^2) + 2gx + 2fy + c = 0$ represents a circle with centre

(A) (B)

(C) (D)

Q19. Equation of latus rectum of the parabola $x^2 = -4ay$ is

(A) (B)

(C) (D)

Q20. $(a - b)\cdot(a + b) =$

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 8 × 2 = 16) 2. Answer briefly any Eight parts from the followings:-

(i) Evaluate $\text{Lim}(x\to-1)\ ((x^3 + x^2)/(x^2 - 1))$

(ii) Define inverse of a function f.

(iii) Show that $x = a \sec\theta$, $y = b \tan\theta$ represent the equation of hyperbola $x^2/a^2 - y^2/b^2 = 1$

(iv) Evaluate $\text{Lim}(n\to\infty)\ (1 - 1/n)^n$

(v) Find f'(x), if $y = x^2 \ln\sqrt{x}$

(vi) Show that $\cos(x + h) = \cos x - h \sin x - h^2/[2 \cos x + h^3/[3 \sin x + \ldots\ldots$

(vii) Determine the interval in which f is decreasing , here $f(x) = \cos x$, $x \in (-\pi/2 , \pi/2)$

(viii) If $x = y \sin y$, Find dy/dx

(ix) Differentiate $\sin^3 x$ w.r.t $\cos^2 x$

(x) If $y = x^4 + 2x^2 + 2$, prove that $dy/dx = 4x\sqrt{(y - 1)}$

(xi) Write the Quotient rule for derivative of two functions.

(xii) Find dy/dx , if $x = at^2$ $y = 2at$

Q3. (Marks 8 × 2 = 16) 3. Answer briefly any Eight parts from the followings:-

(i) Find dy and δy of $y = \sqrt{x}$ x changes from 4 to 4.41

(ii) Evaluate $\int (3 - \cos 2x)/(1 + \cos 2x)\, dx \cos 2x \neq -1$

(iii) Evaluate $\int 1/(x \ln x)\, dx$

(iv) Evaluate $\int (\ln x)^2\, dx$

(v) Evaluate $\int (3x + 1)/(x^2 - x + 6)\, dx$

(vi) Evaluate $\int_0^{(\pi/3)} \cos^2 x \sin x\, dx$

(vii) Find the area between x-axis and curve $y = \sin 2x$ from $x = 0$ to $x = \pi/3$

(viii) Find 'h' such that A(-1 , h) , B(3 , 2) and C(7 , 3) are collinear

(ix) Find 'k' so that the lines joining A(7 , 3) , B(k , -6) and line joining C(-4 , 5) , D(-6 , 4) are perpendicular.

(x) Find point of intersection of lines $3x + y + 12 = 0$, $x + 2y - 1 = 0$

(xi) Find equation of lines represented by $20x^2 + 17xy - 24y^2 = 0$

(xii) Find equation of line through (-4 , 7) and parallel to the line $2x - 7y + 4 = 0$

Q4. (Marks $9 \times 2 = 18$) 4. Answer briefly any Nine parts from the followings:-

(i) Graph the solution set of the linear inequality $3x + 7y \geq 21$ in xy − plane.

(ii) Define feasible region and feasible solution.

(iii) Find an equation of the circle with ends of diameter at (-3 , 2) and (5 , -6)

(iv) Find centre and radius of the circle $4x^2 + 4y^2 - 8x + 12y - 25 = 0$

(v) Find equation of Normal to the circle $x^2 + y^2 = 25$ at $(5\cos\theta , 5\sin\theta)$

(vi) Write equation of parabola with directrix $x = -2$ and focus (2 , 2).

(vii) Find foci and vertices of the ellipse $x^2 + 4y^2 = 16$

(viii) Find equation of Hyperbola with foci $(\pm 5, 0)$ and vertex (3,0)

(ix) Find sum of the vectors $\vec{AB}$ and $\vec{CD}$ given A(1 , -1) , B(2 , 0) , C(-1 , 3) and D(-2 , 2).

(x) let $u = i + 2j - k$, $v = 3i - 2j + 2k$, $w = 5i - j + 3k$. Find $|3v + w|$.

(xi) Find v for which $v \cdot i = 0$, $v \cdot j = 0$, $v \cdot k = 0$.

(xii) Prove that $a \times (b + c) + b \times (c + a) + c \times (a + b) = 0$.

(xiii) Find α so that $\alpha i + j$, $i + j + 3k$ and $2i + j - 2k$ are coplaner.

Q. Section ---------- II

Note: Attempt any three questions. ($10 \times 3 = 30$)

Q5. 5 -(a) If f(x) = { 3x if x ≤ −2 ; $x^2 - 1$ if −2 < x < 2 ; 3 if x ≥ 2 Discuss continuity at x = 2

(b) Differentiate (√x + 1)(x^(3/2) − 1) / (x^(3/2) − x^(1/2)) w.r.t. x

Q6. 6 -(a) Show that ∫ $1/\sqrt{(a^2 + x^2)}$ dx = ln(x + $\sqrt{(a^2 + x^2)}$) + c here a > 0.

(b) If x = sin θ , y = sin mθ , show that $(1 - x^2)y_2 - xy_1 + m^2y = 0$

Q7. 7 -(a) Evaluate the definite integral ∫_{π/6}^{π/2} cos x/(sin x (2 + sin x)) dx

(b) Minimize z = 2x + y subject to the constraints x + y ≥ 3 ; 7x + 5y ≤ 35 x ≥ 0 ; y ≥ 0

Q8. 8 -(a) Find the equation of the tangent drawn from (−7,−2) to $(x+1)^2 + (y-2)^2 = 26$

(b) Using vectors, prove that sin(α − β) = sin α cos β − cos α sin β

Q9. 9 -(a) Find area of region bounded by the triangle whose sides are 7x − y − 10 = 0 , 10x + y − 41 = 0 , 3x + 2y + 3 = 0

(b) Find the centre , foci eccentricity, vertices of ellipse whose equation is $x^2 + 16x + 4y^2 + 16y + 76 = 0$